STRATA: Multistage, Evidence-Grounded Statistical Verification of Temporal Hypotheses in Clinical Timelines

Anonymous Author(s)

ABSTRACT

Language-model systems increasingly claim to test hypotheses against data, but existing benchmarks score only the final answer and hand the system the relevant records already assembled. Over a corpus of clinical timelines, a claim such as "after Moderna, Pain typically occurs before Asthenia" first requires deciding *which reports count*, what the null is and which exact test applies; each decision can be wrong while the verdict is right. Statistical-reasoning benchmarks and hypothesis-testing agents score none of these steps. We present STRATA, a benchmark and staged evaluation framework for verifying temporal hypotheses about clinical event timelines by exact statistical tests. Claims are generated mechanically from the timelines by a template-driven process that enumerates the cores that exist in the corpus, computes every stage of the gold (the plan, the exact evidence sets, the exact p-values and a three-decision verdict with sub-typed abstention) and validates it by mechanical invariants and a deterministic oracle, so that the corpus codebook, the stipulations the labels depend on, is part of the task rather than hidden in the labels. The framework orchestrates verification as three stages, articulate, retrieve, test (ART), each with an output contract, a bounded verify-and-repair loop and a gold counterpart, so a system is scored on its plan, its evidence and its abstention, not the verdict alone, and any stage can be filled with gold to measure the loss of the others. Across five models from a 2024 open 32B pair to current frontier models, the codebook is what frontier models need, and the plan contract is what lets an open model match them with auditable evidence. The temporal conditioning of the denominator is where systems fail, and the verdict hides it.

CCS CONCEPTS

• **Computing methodologies** → **Natural language processing**;
• **Applied computing** → *Health informatics*.

KEYWORDS

claim verification, temporal reasoning, statistical hypothesis testing, benchmark, clinical timelines, text-to-SQL

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1 INTRODUCTION

A growing class of language-model systems promises to test hypotheses against data: given a claim and a dataset, they choose a statistical method, run it and report a conclusion. Benchmarks for this ability [18, 28, 29] score the final answer and hand the system the relevant data already assembled. In most real settings that is the easy half. For the claim *"after Moderna, Pain typically occurs before Asthenia"* over the clinical timelines of adverse-event reports (Figure 1), a system must first decide *which reports count* (a report where the two events share a stage witnesses neither order), *what "no effect" means* (a 50/50 split of the ordered reports) and *which test applies* (a one-sided exact binomial with a minimum-count floor). Each decision is a stipulation about what the sentence means for this corpus, each can be wrong independently, and a verdict can be right while they are wrong: in Figure 1 a one-query baseline counts only the reports that support the claim and still returns the correct verdict.

This paper makes that chain of decisions the object of evaluation, in a setting where every one of them is hard: verifying temporal hypotheses about clinical event timelines by statistical testing. A hypothesis that one event tends to precede another, follow it immediately, come first among several or fall early in the clinical course is a claim about a population of timelines. No single report confirms it; it is verified only by an exact test over the reports whose timelines can witness it, against the null its wording implies. STRATA (Staged Temporal-statistical Reasoning Assessment) is a benchmark and evaluation framework for this task over the clinical timelines of 3,166 vaccine adverse-event reports from MedTempo [12], in which each report's clinical course is a *leveled order* of events. Every instance carries its gold verdict (*supported*, *refuted* or a sub-typed *insufficient*) and the gold of every stage that produced it: the structured plan, the exact report sets in the denominator and numerator, and the exact p-values, all computed mechanically from the timelines and reproduced by a deterministic oracle. We pair the benchmark with an instrumented reference pipeline, ART (articulate, retrieve, test): a stage that articulates the claim as a plan, a retrieval stage and a deterministic test stage, each with an output contract and a bounded verify-and-repair loop. Its stages can be filled with gold, its prompts can state or withhold the corpus *codebook* (the stipulations about how the timelines are to be read, written as text), and its plan can be handed on either as prose or as a *contract*, in which the harness expands each primitive the plan names into that primitive's exact rule. That separates what end-to-end evaluation conflates, the *information* a system is told and the *architecture* it is given.

On a 1,500-claim test set that no prompt decision was made on, with five models from the 2024 open generation to the 2026 frontier, three findings answer three questions: end to end, even frontier models verify the claims only once the corpus's conditioning rules are stated, and agents that write their own analysis do no better than a single query; for open models the contract is what makes the verdict stand on the right evidence, carrying a 31B model to within 0.02 of the frontier at zero API cost; and systems fail at the temporal conditioning of the denominator, deciding which reports

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can witness the claim, a failure that verdict accuracy hides. Our contributions are:

- **Temporal hypothesis verification as statistical testing, with gold at every stage**: a benchmark of hypotheses about the order, adjacency, position and co-occurrence of clinical events across patients' timelines, each verified by a one-sided exact test whose population, null and test follow from the claim's temporal shape; 26 groups and seven families over three difficulty levels, released as a development set, a fact-disjoint test set and a single-report case study, with structured plans, exact evidence sets, exact p-values and a three-decision verdict with sub-typed abstention (§3).
- **A generation process that makes the codebook part of the task**: claims enumerated from the clinical timelines by templates, gold computed mechanically by the exact test for each claim shape, screened against permutation nulls and validated by invariants and an oracle, and the stipulations that turn a temporal sentence into a testable population (which reports can witness an order, how ties and co-existing events count, what chance means for an early event) shipped as text any system can be given (§3).
- **A staged verification method**, Art, that articulates a temporal hypothesis as a plan naming its conditioning rule, null and test, retrieves the conditioned report sets, and decides with a deterministic exact test, so that the verdict is a statistical decision rather than a model judgment; the plan is handed to retrieval as a contract resolved by the harness, each stage has bounded repair, and any stage can be set to gold (§4).
- **A study of five models and two agentic systems** under three research questions, showing that systems fail at the temporal conditioning of the evidence rather than at the verdict, and that open models asked to make the statistical call themselves err on a fifth to two fifths of claims even with exact counts; with stage and repair-budget ablations, case studies and an error analysis of the *insufficient* class (§5, §6).

2 RELATED WORK

Work on language models and statistical hypotheses ranges from choosing a test to running a whole analysis. StatQA [29] asks which test applies, StatFormBench [25] asks for the formulation of a test family, and StatABench [28] scores the final answer of an analysis run with fixed tool functions. P-Bench and Fisher-R1 [18] are closest to our setting: an agent receives a hypothesis with its null stated, a CSV of the relevant data and a glossary, writes R to compute a p-value, and is scored on the decision and the p-value. Agentic systems go further and plan the analysis themselves. LATCH [9] runs LLM-chosen analyses through deterministic R pipelines on EHR data with the method given, VERITAS [23] is a multi-agent co-scientist with a deterministic evidence classifier for image-derived cohorts, and DiscoveryBench and BLADE [11, 17] evaluate open-ended discovery agents against rubrics. Across this line of work the data are either handed over as the relevant subset or left entirely to the agent, the hypotheses concern variables in a table, and only

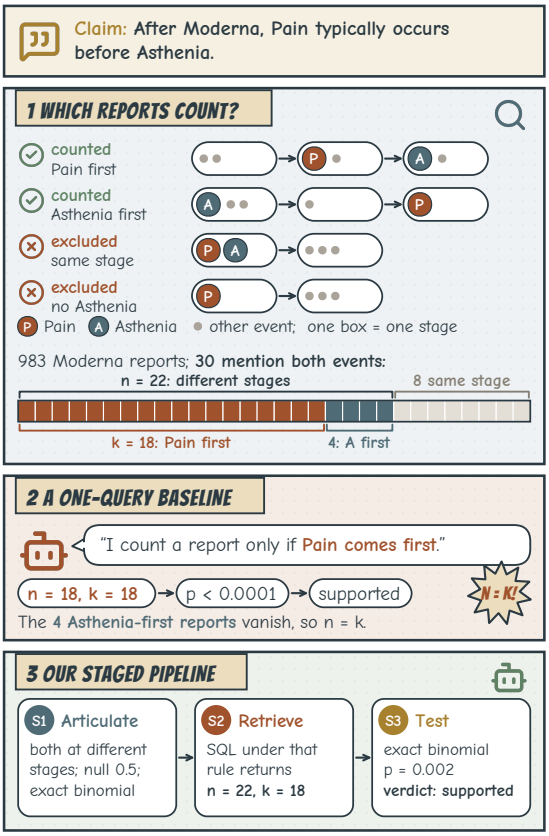


Figure 1: One test claim end to end, with real counts. (1) Four Moderna reports, one per case; the bar splits the 30 reports that mention both events into the denominator n and the numerator k. (2) A one-query baseline and (3) our staged pipeline, both run with the same open model (§5.2, §4).

the final answer or a rubric is scored. None asks the system to decide which records a statistic is computed over, and none releases gold for that decision or for the plan and the test around it.

Temporal reasoning with language models has mostly been evaluated on single texts. Benchmarks ask about the order, duration or frequency of events in a passage or a synthetic event graph [6, 26]; clinical temporal relation extraction labels the order of events within one narrative [2, 24]; and TIMER [4] and CRAFT [12] extract and refine clinical timelines with language models. Clinical question answering over databases likewise asks for one answer per question, with an abstention option in EHRSQL [14] and with a code-writing agent and error-driven retries in EHRAgent [22], and open text-to-SQL models such as OmniSQL [15] and XiYanSQL [16] are built for the same setting. In all of these the answer is a fact about one text, one timeline or one query. A temporal hypothesis is a claim about a population of timelines, and verifying it adds the decisions this paper scores: which timelines can witness the order at all, what chance looks like when events crowd the early stages, and which exact test decides. Strata starts where extraction ends, using the MedTempo timelines released with CRAFT as its data

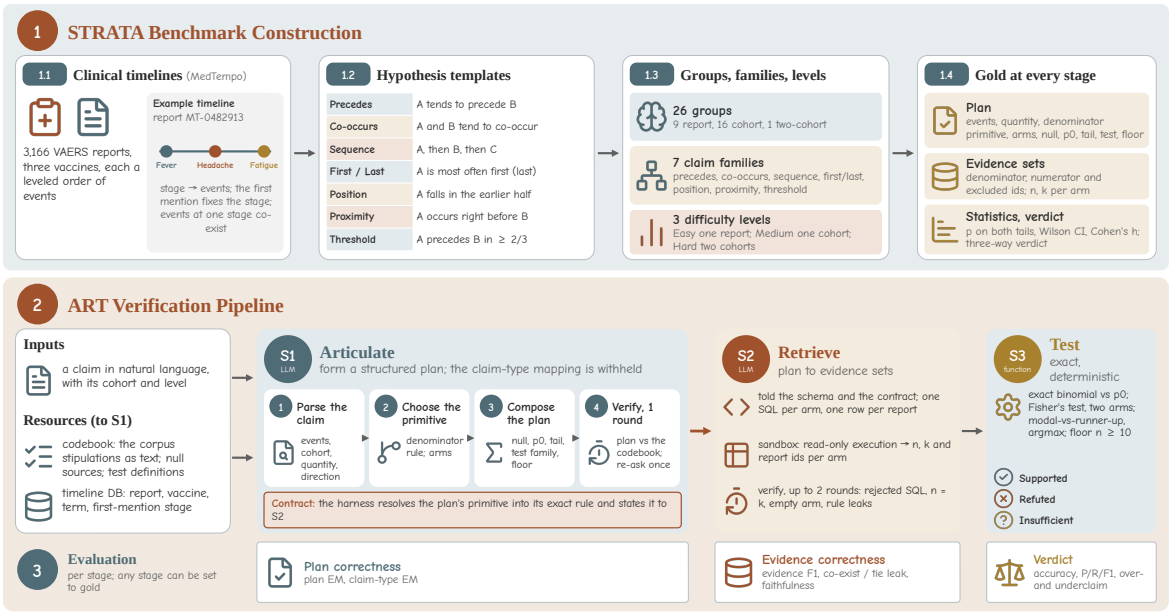


Figure 2: Overview. Top: STRATA benchmark construction, from the clinical timelines (1.1) through the hypothesis templates (1.2) and the groups, families and difficulty levels (1.3) to the gold computed at every stage (1.4). Bottom: the ART verification pipeline, articulate (S1), retrieve (S2) and test (S3), with the contract between S1 and S2 and the verify-and-repair loops; beneath each stage, the measure that scores its output against the gold (3).

source, and adapts EHRSQL, EHRAgent and DiscoveryBench as baselines (§5.2); Table 4 (Appendix A) compares the closest work point by point.

3 THE STRATA BENCHMARK

3.1 Task definition

A record is a temporal hypothesis stated as a claim in natural language, in one of four registers, about a *cohort* (the reports of one vaccine), a *contrast* of two cohorts, or one named *report*, together with the timelines: for every report, its vaccine and the stage at which each event term is first mentioned. The system never sees report text. It must verify the hypothesis by a statistical test over the timelines and return *supported*, *refuted* or *insufficient*. A claim is supported only when its own one-sided tail is significant at $\alpha = 0.05$ and refuted only when the opposite tail is, never for failing to prove it; otherwise it is insufficient, sub-typed as *not-locatable* (no report qualifies), *sparse* (fewer than the minimum count qualify) or *underpowered* (enough qualify but neither tail is significant; the name marks an inconclusive test, not a power calculation).

For a cohort-tier claim the gold is computed in three released steps: (1) a *plan*, giving the events, the quantity compared, the *denominator primitive* (which reports count), the number of arms, the null, the tail and the test family; (2) the *evidence*, the identifiers of the reports in each arm's denominator and numerator; and (3) the exact test with its minimum count. A report-tier claim is one deterministic reading of a report's timeline and carries no test. Figure 1 follows one test claim through these steps.

Several readings of a claim are stipulations about the corpus rather than facts about the sentence, such as whether a report with both events at one stage counts for "precedes" or whether a tie for "first" leaves the denominator. The benchmark ships them as a *codebook* that any system may be given (§4.1) and withholds only the *mapping* from wording to rule and test, which a system is scored on composing.

3.2 Benchmark construction

We use the clinical timelines of MedTempo [12]: 3,166 VAERS adverse-event reports for three COVID-19 vaccines (1,164 Janssen, 1,019 Pfizer, 983 Moderna; 1,660 event terms; 7.2 events and 3.4 stages per report on average), each represented as a *leveled order*, a chain of stages in which each stage holds the MedDRA-coded events reported at that point. A term's *first* mention fixes its stage, and events at one stage *co-exist*, with no order between them. Systems receive the timelines rather than the narratives, so an error is an error of verification rather than of reading, which is the task CRAFT studies.

Hypotheses come from 26 template groups in seven claim families: *precedes*, *co-occurs*, *sequence* of three events, *first/last* among candidates, and, to diversify the null, *position* (an event in the earlier half of a timeline), *proximity* (two events at consecutive stages) and *threshold* (a rate of at least a stated bar). They span three difficulty levels: *Easy* claims read one report's timeline (9 groups), *Medium* claims test a rate within one cohort (16 groups), and *Hard* claims compare two cohorts (1 group). Figure 2 shows one template per family, and Table 5 (Appendix B) lists every group. Each group is rendered in four registers (ORIGINAL, RESEARCH, MEDIA,

CLINICAL) by mechanical slot-filling, with no model-written text. Every decisive core is posed both ways, as a *supported* claim and its *refuted* mirror over identical evidence, so a system that supports both directions of one fact is caught. Each cohort group ships equal thirds of supported, refuted and insufficient, balanced per group so that fixed template wording cannot teach a shortcut from wording to label, and is capped at 600 instances by a seeded draw that keeps mirror pairs together. *Not-locatable* claims are minted from real cohort vocabulary whose joint count is zero, so the claim looks answerable and only querying the timelines reveals the empty denominator.

Gold labels are computed mechanically from the timelines. Each family fixes its denominator, and these rules are the substance of the task: *precedes* counts only reports with both events at distinct stages, whereas *co-occurs* counts every report with both, because co-existence is what it asks about, and *first/last* drops reports in which candidates tie; the catalogue gives every rule. Five kinds of null appear: the symmetric 0.5 for order claims, the named cell against its strongest competitor for argmax claims, equality of two rates for contrasts, a bar read off the wording for threshold claims, and a rate derived from the data for position and proximity claims, which no lookup table can supply. Order and bar claims use a one-sided exact binomial; data-derived nulls use a one-sided binomial at the mean of the per-report chance rates, which is conservative relative to the exact distribution over reports with unequal chances; argmax claims use an exact binomial of the named cell against its strongest competitor, which suffices because beating the strongest competitor implies beating all of them; and contrasts use Fisher's exact test [7]. No χ^2 approximation appears, and the only gate is a minimum denominator of 10 per arm. Each instance releases both one-sided p-values, the Wilson interval [27] and Cohen's h [3], all describing the claim as that instance asserts it.

Before shipping, every group cleared a permutation null matched to its claim shape [10]. Six families of mechanical invariants are checked over the whole enumerated universe of 56,138 instances, and a deterministic oracle that parses each template back into the reference SQL reproduces every gold label, which shows that the labels follow from the stated rules.

The release (Table 1) contains a 1,500-claim test set, fact-disjoint from a 200-instance development set, a 180-claim single-report case study, and the full enumerated pool of 8,763 instances with its generator. Inputs and gold are in separate files, every report identifier is a keyed pseudonym of its VAERS identifier, and a `fact_id` keys every split.

4 METHOD

Our reference pipeline, ART, verifies a temporal hypothesis in three stages: it *articulates* the claim as a plan naming its temporal conditioning rule, null and exact test, *retrieves* the report sets the plan conditions on, and *tests* them with a deterministic exact test, so the verdict is a statistical decision rather than a model judgment (Figure 2). Each stage has an explicit output contract and can be scored against its gold or filled with gold; the two model stages also have a bounded verify-and-repair loop that reads only the contract, the codebook and the data, never the gold. Unless stated otherwise, ART includes these loops.

Test set (Medium / Hard claims)	1,500 (1,418 / 82)
Single-report case study (Easy claims)	180
Development set	200
Groups / families / difficulty levels	26 / 7 / 3
Registers per claim	4
Enumerated pool: instances (records)	8,763 (35,052)
supported / refuted / insufficient	3,521 / 3,521 / 1,721
underpowered / sparse / not-locatable	605 / 498 / 618
Test denominator n, median (IQR)	21 (14–41)
|Cohen's h| on decisive claims, median	0.65

Table 1: The release. Per-group sizes are in the claim catalogue (Appendix B).

4.1 S1: Articulate

From the record (text, level, cohort) the model composes a plan: `events`, `quantity`, `denominator` (a named primitive: *ordered_pair*, *both_present*, *all_present_unique_first*, *consecutive_pair*, *present_offmid*, *presence_split*, *single_report*, ...), `arms`, `null_kind` with p_0 or a reference population, `tail`, `test_family` and a rule for the floor. The prompt states the codebook by situation ("a report with two events tied for earliest has no answer and is excluded"; "order words are distance-blind; only wording that says the two are adjacent conditions on consecutive stages") together with a computed dataset card, the null sources and the test definitions, but withholds the mapping from wording to claim type, so the model composes the plan rather than looking it up; the headline condition is zero-shot, since a worked plan would be a mapping row. The harness derives the claim type from the plan by reverse lookup over its fields, and a plan that names no known shape is *underdetermined* and scored as insufficient. The plan is then checked for consistency with the codebook (a quantity against its denominator, a null kind against its situation, a stated bar against the wording, a valid test family) and re-asked once with the violations named.

4.2 S2: Retrieve

For each arm the model writes one SQL statement over the timeline database (two tables, two views) that returns *one row per report* with four flags: in the denominator, in the numerator, excluded, and an optional cell label for argmax claims. The plan reaches this stage as a *contract*: the harness resolves the plan's claim type and states that type's conditioning rule and arm plan in the prompt, the stipulation for the one primitive the plan names. The alternative, handing over the plan's JSON with the full codebook as *prose* and leaving the model to work out what the primitive means, is the ablation that shows why. The harness executes the statement in a read-only sandbox and aggregates the rows into (n, k) per arm with the identifier sets, so the evidence is auditable against the gold sets. A statement the database rejects, an arm whose denominator equals its numerator, an empty arm beside a full one, and rows that violate the plan's primitive (co-existing reports in an order denominator, ties in a first/last denominator, non-consecutive pairs in an immediate denominator) trigger a re-ask, up to two rounds. The EHRSQL baselines receive the same loop with the checks that need no plan.

4.3 S3: Test

A deterministic function runs the plan's test on the counts (Table 5): the exact binomial against the plan's p_0 (for a data-derived null, p_0 is the mean chance over the retrieved denominator), Fisher's test for two arms, and modal-versus-runner-up for argmax claims. It applies the floor and returns the verdict with both tails. Nothing is inferred at this stage, so it needs no verification.

5 EXPERIMENTS

5.1 Research questions

The experiments answer three questions, each on the test set and each with the same five models in every model slot.

- **RQ1.** How well do current language models verify these temporal hypotheses end to end, as a single query or as an agent, and where do they fail?
- **RQ2.** What does the staged pipeline add over single-query verification for each model, and which stage and which claim families account for the errors that remain?
- **RQ3.** Do correct verdicts rest on correct evidence, and how often do systems decide claims the data cannot decide, or abstain on claims it can?

§6.1–§6.3 answer each in turn; the ablations (§6.4) separate the stages and the repair budget, and the case studies (§6.5) cover single reports and paraphrase.

5.2 Baselines

Three published systems are adapted to the benchmark, all over the same database. *EHRSQL* [14] follows the single-query text-to-SQL protocol: one prompt per claim asks for one query over our schema that marks each report as in the denominator, the numerator or neither, and the harness runs the query and the exact test. *EHRSQL-C* adds the codebook to the prompt, so it differs from Art only in having no plan, no contract and no repair. *EHRAgent* [22] runs as released, writing and executing its own code with up to twelve error-driven retries and four demonstrations, and running its own test; it receives the schema, the exact-test functions and the floors but not the codebook, since its knowledge stage writes the plan itself. *DiscoveryBench*'s reference agent [17] runs as released on the same tables exported as CSV, with the codebook as its domain knowledge, an instruction to end on a verdict line and up to 25 steps; at one to two minutes per claim it runs on a stratified 204-claim subsample (Appendix F).

Stage baselines replace one stage of Art with an established method for that step and set the other stages to gold. For Articulate, a plain prompt gives the model the dataset description, the codebook and the plan schema, without Art's explanations, field rules or verify round, with and without zero-shot chain-of-thought [13]. For Retrieve, OmniSQL [15], fine-tuned from the same base as our 2024 SQL model, writes the SQL from the gold plan. For Test, the StatQA protocol [29] gives the model the gold counts and the test definitions and asks for the test and the verdict. The matching Art rows fill the same stage with Art's own component, and Art's Articulate stage with chain-of-thought is reported as an ablation.

5.3 Evaluation metrics

The verdict is scored as a three-class prediction: accuracy, and precision, recall and F1 macro-averaged over *supported*, *refuted* and *insufficient*, on the cohort tiers (the statistical headline) and on the report tier (a reading task) separately. These four numbers say how often and how evenly a system reaches the right label; they cannot say whether it reached it for the right reason, because a verdict can be right on the wrong evidence and wrong on the right evidence. Because every stage has gold, four evidence measures are therefore reported with the verdict; Figure 4 breaks them down by claim family. *Plan EM* is exact match on the plan's fields. *Evidence F1* is the F1 between the set of reports a system placed in an arm's denominator and the gold set, averaged over arms: it measures whether the system counted the right reports, whatever it concluded from them. *Faithfulness* is the share of correct verdicts whose denominator is exactly the gold set in every arm, that is, the probability that a right answer is right for the right reason. Two abstention measures complete the picture: *overclaim*, the share of gold-*insufficient* claims given a decisive verdict (deciding what the data cannot decide), and *underclaim*, the share of decisive claims answered *insufficient*. We also report the sub-type accuracy of abstentions, the *co-exist leak* (same-stage reports wrongly admitted to an order denominator) and p-agreement with the exact computation.

5.4 Experimental settings

The splits are those of §3.2: the 200-instance development set, on which every prompt decision was made and from which no number enters the main table; the 1,500-claim test set, run once after the prompts were frozen, in the Original register and claim form (the paraphrase case study reruns it in the three other registers); and the 180-claim report case study, kept apart because every capable system reads a single report at 0.98–1.00. Every system is run with each of five models in every model slot: the 2024 open pair Qwen2.5-32B-Instruct (planning) and Qwen2.5-Coder-32B (SQL) [20]; the 2026 open models qwen3.8-27B [21] and gemma4-31B [8], one model for both stages, thinking off; and the frontier models Claude Opus 5.5 [1] and GPT-5.6 [19]. Open models run fully resident on one RTX A5000 with an 8k context, frontier models over their APIs with prompt caching. Verdict accuracies carry 95% bootstrap intervals [5] over instances (1,000 resamples), about ±0.02 on the test set and ±0.01 above 0.98. Per-record cost and latency are in Appendix G; a frontier column on the test set cost $80–89 at list price, and the open models answer a claim in 10–14 seconds on one workstation GPU. Prompts and settings were frozen on the development set before any test run; those tuning analyses are released with the code.

6 RESULTS

Table 2 reports every system, the stage ceilings and the ablations on the same 1,500 claims. The three research questions are answered in turn.

6.1 RQ1: end-to-end verification

Without the codebook, a single query verifies 0.53–0.88 of the claims across the five models (EHRSQL in Table 2); with it, the same query reaches 0.58–1.00 (EHRSQL-C). The gain grows with the model's

test set, 1,500 cohort claims	Qwen2.5-32B pair				qwen3.8-27B				gemma4-31B				Opus 5.5				GPT-5.6			
	Acc	P	R	F1	Acc	P	R	F1	Acc	P	R	F1	Acc	P	R	F1	Acc	P	R	F1
adapted baselines																				
EHRSQL	.529	.555	.528	.518	.587	.605	.587	.582	.635	.647	.634	.629	.878	.882	.878	.878	.876	.877	.876	.876
EHRSQL-C	.575	.604	.574	.570	.748	.764	.747	.748	.859	.885	.858	.861	1.000	1.000	1.000	1.000	.984	.984	.984	.984
EHRAgent	.605	.694	.606	.600	.839	.863	.839	.843			–				–		.792	.808	.792	.794
DiscoveryBench†			–		.647	.812	.647	.711	.706	.784	.706	.729			–				–	
adapted stage baselines (other stages at gold)																				
Articulate: ART	.953	.985	.953	.969	.917	.942	.917	.928	.992	.992	.992	.992	.993	.993	.993	.993	.993	.993	.993	.993
ART + chain-of-thought		*running*				*running*			.995	.995	.995	.995			–				–	
plain prompt		*running*				*running*				*running*					–				–	
plain prompt + chain-of-thought		*running*				*running*				*running*					–				–	
Retrieve: ART	.836	.879	.835	.840	.944	.945	.944	.944	.989	.990	.989	.989		≥.999 on all four				≥.999 on all four		
OmniSQL-32B‡	.772	.814	.771	.775		–				–					–				–	
Test: ART (exact test)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
StatQA	.611	.706	.609	.575	.619	.784	.617	.539	.777	.816	.776	.776	.999	.999	.999	.999	.999	.999	.999	.999
ablations of ART																				
− verifier	.761	.845	.760	.777	.735	.815	.734	.744	.844	.886	.843	.848	.999	.999	.999	.999	.999	.999	.999	.999
− exact test (model verdict)	.512	.651	.510	.482	.568	.612	.566	.503	.770	.809	.769	.769			–				–	
− contract (plan as prose)	.663	.704	.663	.671	.841	.870	.841	.851	.947	.948	.947	.947			–				–	
ours																				
ART	**.810**	**.876**	**.809**	**.827**	**.872**	**.895**	**.872**	**.882**	**.981**	**.982**	**.981**	**.981**	**.993**	**.993**	**.993**	**.993**	**.983**	**.983**	**.983**	**.983**

Table 2: Main results on the test set. Under each model: verdict accuracy (Acc) and macro precision, recall and F1 (P, R, F1) over the three verdict classes, leading zeros dropped; only the model changes across column groups. Intervals on accuracy are ±0.02 (0.6–0.9) and ±0.01 (above 0.98). Every baseline is adapted to STRATA: it reads the same timeline database and returns the three-way verdict, and its name identifies the protocol or released system, not its original setting (§5.2). ‡Fine-tuned from Qwen2.5-Coder-32B, so listed under the 2024 pair. †Run on a stratified 204-claim subsample (Appendix F), with the two 2026 open models. A dash marks a run not made.

ability to apply the text: 0.05 for the 2024 pair, 0.16 for qwen3.8, 0.22 for gemma4 and 0.11–0.12 for the frontier models, which then verify 0.98–1.00 of the claims with every denominator but one exact. The task is therefore not hard for a frontier model that has the stipulations; it measures whether they are known and applied. The 2024 pair cannot apply them: given the codebook it stops admitting most co-existing reports into order denominators but starts writing queries that define the eligible set as the supporting set ($n = k$, 325 claims), and gains almost nothing.

Every error a model makes without the codebook is a stipulation about which reports can witness a temporal claim, and the attribution of Appendix H names it per claim. The frontier models' errors fall on four such rules: first/last denominators that admit reports missing one of the candidates (102–150 claims, first/last accuracy 0.66–0.74); adjacency counted across co-existing events (35–63 claims, proximity 0.79–0.90); position claims over reports that have no earlier half (up to 31); and order and co-occurrence denominators that admit reports with only one of the events. The open models make the same errors in larger numbers and add the one the codebook states first, co-existing reports in order denominators (281–325 claims). None of these is a failure of SQL: the queries run and count the wrong reports.

The agents that write and run their own analysis do no better. Without the codebook, EHRAgent's retries help the models that also fail the single query, by 0.08 for the Coder and 0.25 for qwen3.8, but cost GPT-5.6 0.08 (0.792 against 0.876): free to choose, it keeps ties in first/last denominators, applies multiple-comparison corrections and counts co-occurrence over reports with either term, each a defensible convention that is not this corpus's. Given the codebook, DiscoveryBench's agent scores below the single query with the same model on the same 204 claims: DiscoveryBench's agent 0.65 and 0.71 against 0.76 and 0.88 for qwen3.8 and gemma4 (Appendix F, where every other system scores within its own interval on the same claims). An agent that writes its own denominator and test has more ways to depart from the stipulations it was given, and nothing checks that it did not.

6.2 RQ2: the staged pipeline and where errors remain

The frontier models have no verdict headroom with the codebook: they score the same under ART as end to end, and the verifier costs them a little, because its checks flag a few correct $n = k$ statements and the pipeline abstains rather than guess. What the pipeline gives them is not accuracy but an auditable trail: a plan whose fields are scored, a report set per arm, and a test the harness ran, which the one-prompt query does not produce even when it is right. For the open models the contract is what makes the verdict stand on the right evidence: with the contract and the verifier every open model gains between 0.12 and 0.24 in verdict accuracy over EHRSQL-C, and gemma4-31B lands within 0.02 of the frontier at zero API cost. The contract does not make the models better at SQL; it removes the interpretation step on which they were failing, and §6.3 shows the same gain on the evidence.

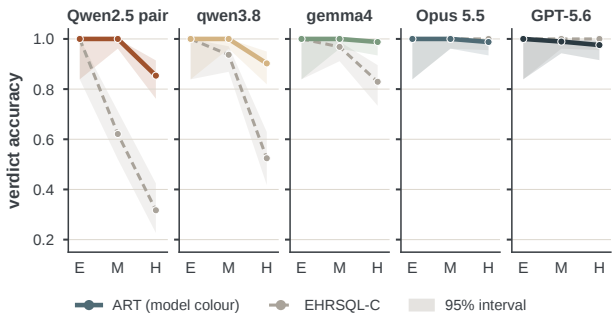


Figure 3: Verdict accuracy on one claim shape, *A precedes B*, at the three difficulty levels: E, read off one report (P-R1, 20 claims); M, tested in one cohort (P-V1, 95); H, compared across two cohorts (P-M2, 82). One panel per model: ART against EHRSQL-C, which shares its model, codebook, database and exact test but has no plan, contract or repair. Bands are 95% Wilson intervals over claims.

Accuracy under ART varies by claim family, and the families do not fail together (Figure 6, Appendix C). The 2024 pair's losses sit in two families whose rule it cannot carry out: first/last, an argmax over three candidates with ties dropped from the denominator, and threshold, where one group states its bar in the text. The 2026 open models clear first/last and each misses a different family, qwen3.8 on position and on the precedes variant whose null is a data-derived rate, gemma4 only on proximity and threshold. Threshold is the one family where the frontier models also drop, and their planner-alone rows drop by the same amount, so the loss is in composing the bar into the plan, not in retrieval.

The three levels contain different families, so their pooled accuracies mix difficulty with family; Figure 3 holds the claim shape fixed instead and compares ART with EHRSQL-C, which differs from it only in the plan, the contract and the repair. Difficulty is real: under ART every model verifies the single-report and one-cohort claims, and the open models lose 0.15 (pair), 0.10 (qwen3.8) and 0.01 (gemma4) when a second cohort adds a second denominator and a two-sample test. EHRSQL-C loses far more over the same steps, 0.68, 0.48 and 0.17, the pair already 0.38 at the one-cohort step, while the frontier models stay at or near 1.0 in both. The contract does not remove the cost of the harder level; it cuts it by a factor of five or more for the open models. The Hard level has 82 claims, so its intervals are about ±0.08.

6.3 RQ3: evidence and abstention

Every prediction records the reports it counted, so each verdict can be checked against the gold sets. End to end, verdict accuracy overstates what the systems computed: the one-prompt query with the codebook rests 40% (pair), 69% (qwen3.8) and 86% (gemma4) of its right verdicts on the exact denominator, so the pair's 0.575 accuracy is 0.23 right for the right reason. Figure 1 (panel 2) shows one such case from qwen3.8: the query admits only the reports that support the claim, so $n = k$ and the verdict is *supported* whatever the data say. Under the contract the same shares are 86%, 97% and 94%, a larger gain than in accuracy and largest for the weakest model, and the frontier models are exact on all but a handful of claims in both systems. The agents cannot be checked at all: EHRAgent and DiscoveryBench's agent return a verdict from their own code, so a right verdict from them says nothing about which reports it counted.

Figure 4 breaks ART's evidence down by claim family. Almost every cell is at or near zero error for gemma4 and the frontier models and for most families of the 2026 open models, and overclaim stays at a few percent everywhere. The dark cells are the failures that remain, and each has a cause the plan exposes: the 2024 pair's first/last row, where it retrieves the wrong set and abstains on four claims in five rather than decide; qwen3.8's overclaim on first/last; and the frontier models' underclaim on threshold, the H-V5 slip discussed below. One failure is invisible in the verdicts: gemma4 is right on 0.97 of proximity claims, but a fifth of those right verdicts rest on a denominator that admits non-adjacent pairs. EHRSQL-C is weaker on the same families, with faithfulness of 0.06–0.41 on first/last for every open model.

The three-decision verdict can decide what the data cannot decide or abstain on what it can, and Figure 5 separates the two. Overclaim is the share of the 505 gold-*insufficient* claims (176 *not-locatable*, 157 *sparse*, 172 *underpowered*) given a decisive verdict. Underclaim is the share of the 995 decisive claims answered *insufficient*, split by what the prediction shows, which is also the sub-type the system claimed: an empty denominator claims *not-locatable*, one below the floor *sparse*, a testable one *underpowered*. EHRAgent returns a bare verdict, so its causes cannot be observed.

Overclaim comes mostly from the agents and the one-prompt baselines. EHRAgent with the Coder decides more than half of the undecidable claims and nearly all *underpowered* ones, because it runs its own test. EHRSQL-C overclaims mostly on *underpowered* claims, where a wrong denominator moves the tail, and fills *not-locatable* denominators with reports that do not qualify. ART overclaims on a few percent at most, because the exact test, not the model, abstains, and its abstentions name the right sub-type on 0.87–1.00 of claims against 0.79–1.00 for EHRSQL-C.

Underclaim is concentrated in the open models. The one-prompt pair abstains on nearly four decisive claims in ten, mostly on a wrong denominator or one below the floor; the stronger EHRAgent models trade overclaim for underclaim rather than for correct verdicts; under ART the pair's abstentions are mostly queries that still fail after two repairs. The frontier models' few abstentions all fall in one group, H-V5, where the plan reads a stated bar as an argmax claim and runs the wrong test on the right counts.

6.4 Ablations

The ablations in Table 2 remove one part of ART at a time: the verifier's repair loops; the exact test, replaced by the model's own verdict from the pipeline's counts; and the contract, the plan handed to S2 as prose with the codebook. Removing both the plan and the contract leaves EHRSQL-C; everywhere else ART includes the verifier.

The stage rows in Table 2 fill one stage at a time with the model and leave the others at gold, so each row is the loss attributable to that stage alone. Planning is nearly solved by every model: with

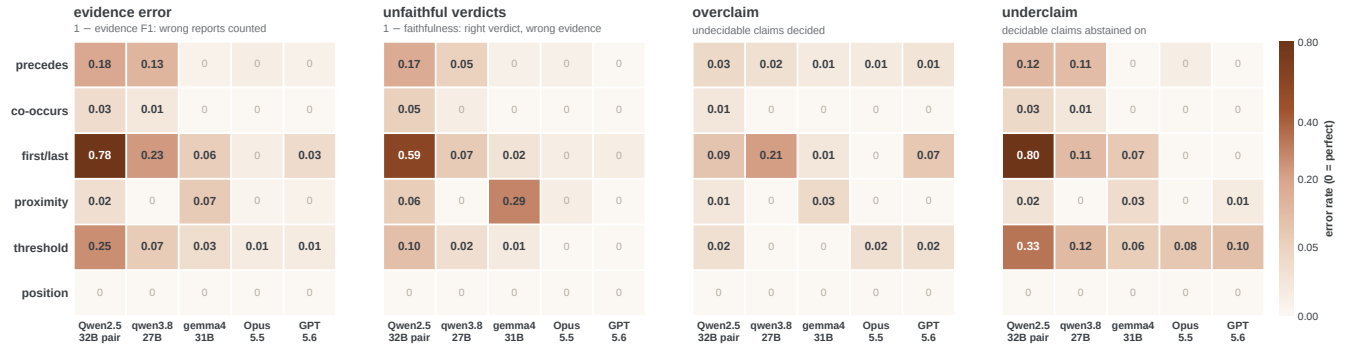


Figure 4: The four evidence measures of Art on the test set, by claim family and model, drawn as error rates so that 0 is perfect in every panel: the share of wrongly counted reports (1−evidence F1), of right verdicts resting on the wrong evidence (1−faithfulness), of undecidable claims decided (overclaim) and of decidable claims abstained on (underclaim). The colour scale is square-root, so small errors remain visible.

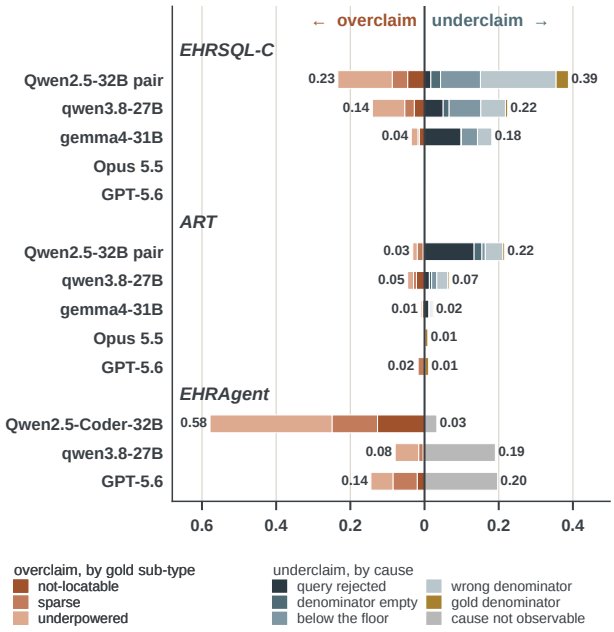


Figure 5: The *insufficient* class on the test set, one row per system and model. Left: overclaim, by gold sub-type. Right: underclaim, by cause; grey where the cause is not observable. Numbers are row totals.

the mapping withheld and the codebook given, the plan is composed right on at least nine claims in ten, and the gemma4 planner matches the frontier. Retrieval is where the 2024 pair loses, its Coder dropping a sixth of the claims even with the gold plan, so the full pipeline compounds planner errors on a stage that was already failing. The 2026 models retrieve well with the gold plan, so their remaining loss is the plan, and gemma4 under Art sits within 0.01 of both its ceilings. The test stage is a function the open models cannot compute: under the StatQA protocol on the exact counts the open models are wrong on a fifth to two fifths of claims, and replacing the exact test with the model's verdict on the pipeline's own counts loses 0.2 to 0.3.

The hand-off between S1 and S2 is what the stage rows cannot see, because Retrieve receives the gold plan already resolved. The − *contract* row hands the same composed plan to S2 as prose with the codebook, so the plan, the model and the repair rounds are unchanged and only the resolution of the named primitive into its rule is removed. This costs the 2024 pair 0.15 of verdict accuracy (0.810 to 0.663) and the 2026 models 0.03 (0.872 to 0.841 for qwen3.8, 0.981 to 0.947 for gemma4). Without the contract the pair is barely above the one-prompt baseline with the same codebook and the same two repair rounds (0.638), while the newer models keep most of the decomposition's gain (EHRSQL-C: 0.748 and 0.859): the weaker the model, the more of the pipeline's gain the contract carries. The development-set

Figure 8 (Appendix E) runs the pipeline with zero to three verify-and-repair rounds. For the 2026 open models the first round does almost all of the work, about +0.13 in accuracy with evidence F1 rising with it, and the next two rounds add a few thousandths; the 2024 pair gains little per round because its Coder resubmits the same mistake when told what was wrong. At the frontier every round costs a little, since the checks fire on statements that are already correct and the pipeline turns a right verdict into an abstention. Two rounds is the operating point for every open model; a model that does not need the verifier is not helped by it.

6.5 Case studies

The report tier asks a system to locate one report by its pseudonymous identifier and read an ordering from its timeline; the plan has one arm and the test stage is a deterministic rule. Every system with a 2026 model does this at 0.98–1.00 with or without the contract (Table 3); the 2024 Coder end to end does not, and reaches 1.000 only under the contract. The level is trivial as statistics and still separates systems that read what the contract asks them to read from systems that do not.

Every test claim also exists in three paraphrase registers (research, media, clinical), and the case study reruns the three open

180 single-report claims	Qwen2.5 pair	qwen3.8	gemma4	Opus 5.5	GPT-5.6
EHRSQL	0.433	0.989	0.744	—	—
EHRSQL-C	0.750	1.000	1.000	1.000	1.000
Articulate: ART (Retrieve, Test at gold)	1.000	1.000	1.000	—	—
ART	1.000	1.000	1.000	0.983	1.000

Table 3: The report-tier case study: verdict accuracy on the 180 single-report claims.

models on all of them (Figure 7, Appendix D); the open models are the ones with headroom to lose, and the frontier models, at 0.98–1.00 on the original wording, would show little, so they are left to future work with free paraphrase. The research register costs the most, 0.05 to 0.13 of accuracy, the media and clinical registers less (at most 0.07). Rerunning the planner alone on the same registers (not shown) loses almost exactly what the full pipeline loses for the 2026 models, so the cost of paraphrase is paid at planning and passes through the contract unchanged; the 2024 pair's pipeline loses less than its planner, because its retrieval stage rather than its plan is the bottleneck. gemma4 stays ahead in every register, while qwen3.8 and the pair swap places on the research register, where qwen3.8's planner loses most. The registers are templated.

7 CONCLUSION

STRATA evaluates temporal hypothesis verification as a chain of scored decisions (which reports can witness a claim, what the null is, which exact test applies, what the evidence was) and shows that systems fail at the first and that the verdict hides it. Frontier models verify the claims once the codebook is stated; open models need the plan to be a contract, and with it a 31B model on one GPU reaches the frontier's accuracy with an auditable denominator, while models asked to make the statistical call themselves err even on exact counts. Benchmarks of this kind should publish their codebook with their data and score the evidence with the answer.

The study has four limits. It uses one corpus, so the codebook and the primitive definitions are specific to MedTempo, and porting the framework means writing them for another corpus. The claims are generated from templates over raw MedDRA terms, so their wording varies only across the four registers, and the oracle shows that labels follow from the stated rules, not that free-form wording would carry them. Decisive labels carry false positives at the α rate by construction. And the patterns are statistical regularities in spontaneous reports, not causal findings.

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A POSITIONING AGAINST RELATED WORK

Table 4 sets the closest benchmarks and systems against STRATA along the decisions of §2.

B CLAIM CATALOGUE

Table 5 lists all 26 template groups: the claim as it is worded in the original register, which reports count toward its denominator, the null, the exact test and the size of the released pool. Every population test is one-sided in the claim's direction at $\alpha = 0.05$, with a floor of $n \geq 10$ per arm below which the verdict is *sparse* ($0 < n < 10$) or *not-locatable* ($n = 0$). Each template also exists in three paraphrase registers.

C ACCURACY BY CLAIM FAMILY

Figure 6 gives ART's verdict accuracy by claim family for each model, the data behind the claim-family discussion in §6.2.

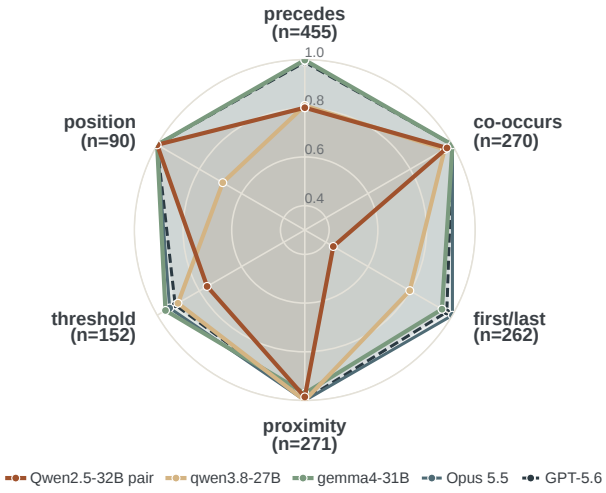


Figure 6: Verdict accuracy by claim family on the test set, our pipeline with the verifier, one polygon per model; rings at 0.4, 0.6, 0.8 and 1.0, family sizes in parentheses.

D PARAPHRASE REGISTERS

Figure 7 gives ART's verdict accuracy in each paraphrase register for the three open models, the data behind the paraphrase case study in §6.5.

E VERIFY-AND-REPAIR ROUNDS

Figure 8 shows what each verify-and-repair round adds to ART's verdict accuracy, the data behind the repair-round discussion in §6.4.

F THE AGENTIC SUBSAMPLE: EVERY SYSTEM ON THE IDENTICAL 204 CLAIMS

DiscoveryBench's agent takes one to two minutes per claim, so it was run on 204 claims of the test set (12 per cohort group, four per

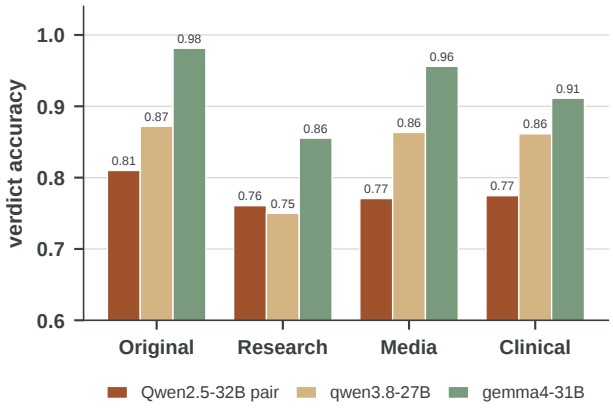


Figure 7: Verdict accuracy by paraphrase register on the test set for the three open models under ART. Original is the main test run; the other registers rerun the same 1,500 claims.

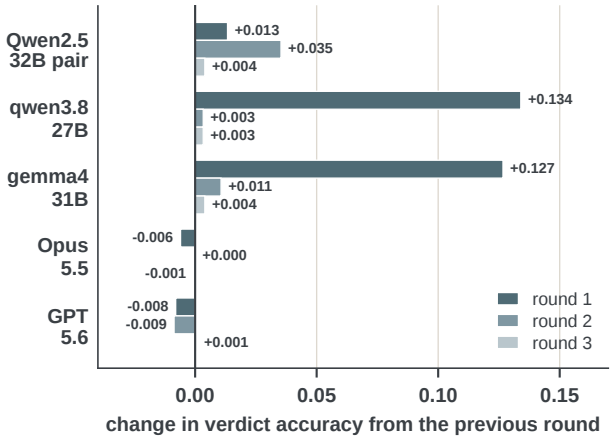


Figure 8: What each verify-and-repair round adds: the change in verdict accuracy from the previous round, for rounds one to three, on the test set; gains extend to the right, losses to the left.

verdict, the test set's own stratification; selected before any run). Table 6 scores the other systems on the same 204 claims next to their full-set numbers, with 95% bootstrap intervals over claims (2,000 resamples).

G COST AND LATENCY

Table 7 gives, per test record, the wall time of the sequential harness, the tokens exchanged and the dollar cost from the providers' own usage counts at list price with prompt caching; local rows cost GPU time only. The contract pipeline makes two calls per record where the Straight-to-SQL baseline makes one, so it exchanges about twice the tokens, and the verifier adds 10–20% on top. A full ten-row frontier column on the test set cost $80 (Opus) and $89

Work	Null	Test selection	Test execution	Event-order claims	Clinical data	Gold released
benchmarks						
StatQA [29]	❍ applicability only	✓ model picks from a catalogue, scored	✗ none run	✗ no	✗ no	✗ final answer
StatFormBench [25]	❍ test family	❍ model names the family	✗ none run	✗ no	✗ textbook	❍ formulation
StatABench [28]	❍ test choice	✓ model picks, scored	❍ fixed tool functions	✗ no	✗ no	✗ final answer
P-Bench / Fisher-R1 [18]	✗ given	❍ model picks, unscored	❍ model-written R	❍ survival only	❍ partly	❍ reference p
DiscoveryBench / BLADE [11, 17]	❍ open-ended	❍ agent, rubric-scored	❍ agent code	✗ no	✗ no	❍ rubric
EHRSQL [14]	✗ none (QA)	✗ none	✗ SQL; no test	✗ no	✓ EHR	❍ SQL, answer, abstention
systems						
EHRAgent [22]	✗ none (QA)	✗ none	✗ agent code; no test	✗ no	✓ EHR	– system
LATCH [9]	✗ given	✗ given by the source study	✓ deterministic R pipelines	✗ no	✓ NHANES	– system
VERITAS [23]	❍ planned by agent	❍ agent plans	❍ own code + rule	✗ no	✓ imaging	– system
Strata + Art (this work)	✓ **composed**: kind, p_0, population	✓ **derived from the plan** (null, arms); scored by plan EM	✓ **harness**: three exact tests, floor	✓ **7 families**	✓ **VAERS timelines**	✓ **plan, evidence, p, verdict**

Table 4: Positioning against related work (a comparison of what each work asks and scores, not of results). *Null*: who states the null and the population it is tested over. *Test selection*: who chooses the test and whether that choice is scored. *Test execution*: what computes the statistic the verdict rests on, and with which tests. *Event-order claims*: hypotheses about the order or timing of events within a record. *Gold released*: what a benchmark scores besides the final answer; systems release no gold of their own. Marks: ✓ has it, ❍ partly, ✗ lacks it, – not applicable.

group	claim template (original register)	reports counted	null	exact test	inst.	med. n
read off one report (Easy)						
P-R1	In report r, A occurred before B.	the named report	–	none	400	–
P-R2	In report r, A came first, before B.	the named report	–	none	400	–
C-R1	In report r, A and B co-occur.	the named report	–	none	400	–
S-R1	In report r, A, B, C occur in the order $A \to B \to C$.	the named report	–	none	400	–
F-R1	In report r, A occurs first among A, B, C.	the named report	–	none	400	–
F-R2	In report r, A occurs last among A, B, C.	the named report	–	none	400	–
F-R3	In report r, A is the first of A, B, C to occur.	the named report	–	none	400	–
T-R1	In report r, A occurred in the first half of the timeline.	the named report	–	none	400	–
Y-R1	In report r, A occurred in the stage immediately before B.	the named report	–	none	400	–
tested in one cohort (Medium)						
P-V1	After v, A typically occurs before B.	both events, at distinct stages	0.5	binomial	501	17
P-V3	After v, A usually comes before B.	both events, at distinct stages	0.5	binomial	501	17
P-V2	After v, B follows A more often than it follows C.	both events, at distinct stages	two rates equal	Fisher	600	22
P-V4	After v, A precedes B more often than C precedes D.	both events, at distinct stages	two rates equal	Fisher	600	24
C-V1	After v, A and B typically co-occur.	both events present	0.5	binomial	600	19
C-V2	After v, A and B occur together more often than their separate rates would predict.	whole cohort, split on presence of A	two rates equal	Fisher	600	229
C-V3	After v, co-occurrence is the dominant relation for A and B.	both events present	runner-up	modal vs. runner-up	522	23
F-V1	After v, A is most commonly the first event among A, B, C.	all candidates present, a unique first	runner-up	modal vs. runner-up	102	14
F-V2	After v, A is most commonly the last event among A, B, C.	all candidates present, a unique last	runner-up	modal vs. runner-up	138	14
F-V3	After v, A is the most common first event among A, B, C.	all candidates present, a unique first	runner-up	modal vs. runner-up	102	14
T-V1	After v, A appears in the earlier half of the timeline more often than chance would place it there.	A present, off the middle stage	data-derived	binomial	102	145
Y-V1	After v, when A and B fall in consecutive stages, A usually comes first.	both events, at consecutive stages	0.5	binomial	204	17
Y-V2	After v, in reports where A and B are in consecutive stages, A comes first more often.	both events, at consecutive stages	0.5	binomial	204	17
Y-V3	After v, A and B fall in consecutive stages more often than chance would give.	both events, at distinct stages	data-derived	binomial	105	26
H-V1	After v, A tends to precede B in at least two out of three cases where their order is recorded.	both events, at distinct stages	bar in text, 2/3	binomial	105	16
H-V5	After v, A tends to be the first of A, B and C in at least half of cases.	all candidates present, a unique first	bar in text, 0.5	binomial	66	13.5
compared across two cohorts (Hard)						
P-M2	A precedes B more often after v_1 than after v_2.	both events, at distinct stages	two rates equal	Fisher	87	21.5

Table 5: Claim catalogue: the 26 template groups by difficulty level. A–D are event terms, v a vaccine cohort and r a report; for example, P-V1 is realised as "After Pfizer, Headache typically occurs before Fatigue". *Reports counted*: the denominator rule. *Null*: 0.5 is the symmetric null; *runner-up* tests the named candidate against its strongest competitor; *two rates equal* compares two arms; *data-derived* computes chance from the denominator reports (the structural early rate, or $2/L$ for adjacency); a *bar in text* is read from the wording. *inst.*: released instances; *med. n*: median test denominator. Report-tier claims are read off one timeline with no test.

system	full test set (1,500)	same 204 claims
DiscoveryBench agent + codebook (qwen3.8-27B)	—	0.647 [0.583, 0.711]
DiscoveryBench agent + codebook (gemma4-31B)	—	0.706 [0.642, 0.770]
EHRAgent, own test (Coder-32B)	0.605 [0.579, 0.629]	0.559 [0.490, 0.623]
EHRAgent, own test (qwen3.8-27B)	0.839 [0.820, 0.857]	0.868 [0.819, 0.912]
EHRAgent, own test (GPT-5.6)	0.792 [0.771, 0.812]	0.784 [0.730, 0.838]
EHRSQL-C (Qwen2.5-32B pair)	0.575 [0.549, 0.599]	0.593 [0.525, 0.662]
EHRSQL-C (qwen3.8-27B)	0.748 [0.725, 0.769]	0.755 [0.696, 0.809]
EHRSQL-C (gemma4-31B)	0.859 [0.839, 0.875]	0.877 [0.833, 0.922]
Art (Qwen2.5-32B pair)	0.810 [0.791, 0.830]	0.804 [0.750, 0.858]
Art (qwen3.8-27B)	0.872 [0.855, 0.889]	0.887 [0.843, 0.931]
Art (gemma4-31B)	0.981 [0.974, 0.988]	0.980 [0.961, 0.995]
Art − verifier (Opus 5.5)	0.999 [0.997, 1.000]	1.000 [1.000, 1.000]

Table 6: Verdict accuracy on the 204-claim agentic subsample against the full test set, same systems, same claims.

(GPT-5.6, which spends hidden reasoning tokens on the test-stage judgment); EHRAgent with GPT-5.6 cost $56 for its one row (4,501 calls, 11.3M prompt tokens of which 5.0M cached, 1.4M completion tokens, reasoning off); a 31B open model answers a test claim in 14 seconds on one workstation GPU and a 27B one in 10.

system	model	s / record	tokens / record	USD / record
D′	Opus 5.5	6.0	8,900	0.014
D′	GPT-5.6	5.1	5,500	0.011
D′	Qwen2.5-32B pair	12.3	5,000	0
D′	qwen3.8-27B	10.1	5,100	0
D′	gemma4-31B	14.1	5,300	0
B	Opus 5.5	4.6	4,600	0.011
B	GPT-5.6	5.5	3,000	0.009
B	Qwen2.5-Coder-32B	2.8	2,850	0
B	qwen3.8-27B	6.2	3,000	0
B	gemma4-31B	6.1	3,100	0

Table 7: Cost and latency per record (test set for the local models, usage counts from the test runs for the frontier models). Tokens are prompt plus visible completion. Local latencies are measured on one RTX A5000 with the model fully resident; the Qwen2.5 pair's includes swapping the two models between stages.

H RETRIEVAL FAILURES BY THE RULE VIOLATED

Every prediction records the report identifiers it placed in each arm, so every wrongly included or dropped report can be checked against the claim's rule in the database and named. Table 8 attributes each test-set record with a wrong or missing denominator to its dominant cause, for the one-prompt baselines, the repaired baseline and Art with and without the verifier, and gives the faithfulness score each column implies. Generated by `scripts/test_attribution_table.py` without model calls; columns for runs still in progress are added as they finish.

	EHRSQL	EHRSQL-C					EHRSQL-C + repair		Art − verifier	Art				
dominant cause per record	pair	pair	qwen3.8	gemma4	Opus	GPT	pair	gemma4	pair	pair	qwen3.8	gemma4	Opus	GPT
no plan from S1 (S2 not run)	0	0	0	0	0	0	0	0	48	47	37	0	0	0
SQL failed to execute	188	26	77	152	0	0	3	0	250	166	2	11	0	0
denominator = numerator ($n = k$)	75	325	322	46	0	0	45	45	48	30	27	1	0	0
event absent from the report	412	350	149	97	0	0	275	98	168	91	72	0	3	10
report from the wrong cohort	42	0	2	0	0	0	1	0	40	23	6	0	0	0
co-exist leak into an order denominator	325	176	20	0	0	0	178	0	32	17	4	0	0	0
gapped pair in an *immediate* denominator	92	11	13	22	0	0	12	22	0	17	0	83	1	1
unsplittable report in a position denominator	14	0	0	0	0	0	70	0	0	0	0	0	0	0
tie leak into a first/last denominator	14	79	86	24	0	0	93	33	1	12	0	1	0	0
presence split narrowed to co-occurrence	63	76	14	0	0	0	78	0	0	0	0	0	0	0
co-exist report wrongly dropped	1	7	0	0	0	0	7	0	0	0	0	0	0	0
other omission	11	36	18	49	0	1	87	89	29	43	28	9	1	0
other over-inclusion	0	0	1	0	0	0	0	0	0	0	0	0	0	0
wrong or missing denominators	1237	1086	702	390	0	1	849	287	616	446	176	105	5	11
of which verdict still right	523	504	317	127	0	0	379	139	180	119	36	79	3	1
faithfulness	0.26	0.40	0.69	0.86	1.00	1.00	0.60	0.90	0.77	0.86	0.97	0.94	1.00	1.00

Table 8: Retrieval failures on the test set (1,500 cohort-tier claims) attributed to the conditioning rule violated. A record takes its dominant cause; "SQL failed to execute" counts records whose statement the sandbox rejected after every repair round; "denominator = numerator" is a query that defines the eligible set as the supporting set. The last rows give the number of records with a wrong or missing denominator, how many of those still returned the right verdict, and faithfulness, the share of right verdicts resting on the exact gold set.